

DUSTBADGE · REQUIREMENTS

DustBadge requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002): R2 restated to 1 mg/m ³ with over-range flag, R3 reference rule, R11 sun use rule; statuses from DBG-CAL-001 v0.2 with the 2,000 mAh cell
0.3	2026-09-25	Amish Chadha	TRL 3, status from DBG-CAL-001 for every requirement; R8 restated for the inlet; R6 defaults follow DBG-DDR-001 D3; R11 and R3 now not met, R10 not verifiable at TRL 3
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

DustBadge requirements

These are the requirements for the concept, with their status from the TRL 3 calculations in DBG-CAL-001 v0.2. On 2026-09-25 Amish accepted the review recommendations (DBG-DDR-002): R2 is restated to the sensor's 1 mg/m³ range with an over-range flag, R3 keeps $\pm 25\%$ but requires a larger reference sample at low-dust, high-silica sites, R11 adds a use rule for full sun, and the cell grows to 2,000 mAh. On paper two requirements are not met (R14 intrinsic safety, out of scope, and R15 cost, \$91.00 against \$90), two are at risk (R3 accuracy and R11 site conditions), one cannot be verified at TRL 3 (R10), and ten are met, six by calculation and four by design. The budget figure for the larger cell is still open, awaiting Amish (DBG-DDR-002, O3).

Table 1. Requirements and concept status.

ID	Requirement	Target	Status at TRL 3 (DBG-CAL-001)	Verification (TRL 3 or later)
R1	Measure the respirable dust fraction	Mass concentration of particles up to 4 µm (PM4), a proxy for the ISO 7708 respirable convention	Met by design. PM4 is a proxy: for quartz, optical PM4 would include grains up to about 6.5 µm aerodynamic, while the sensor undersees coarse grains (CAL-001, D)	Datasheet review; later comparison with a cyclone sampler
R2	Cover the working range	0 to 1 mg/m ³ respirable dust; minutes above the range flagged, logged at the range limit and counted toward the projection. Restated from 0 to 5 mg/m ³ for the first sectors in D6 (DBG-DDR-002 D11)	Met on paper. The RCS limit lies in range only for silica fractions of 5 % or more; above the range the badge reports only that dust was over range (CAL-001, E)	Calculation of expected site ranges; chamber test on CalRig
R3	Accuracy after site calibration	Shift average within ±25 % of a co-located gravimetric sample. Where respirable dust at the action level is below about 100 µg/m ³ (low-dust, high-silica sites such as quartz-rich stone), the reference is a 4.2 L/min cyclone over two shifts (DBG-DDR-002 D9)	At risk. Met on paper with a factor per task on the same badge: ±21 % (quarry), ±22 % (quartz-rich stone with the D9 reference). Not met with one factor shared across badges per site: ±29 to ±30 % (CAL-001, F). Error terms assumed	Error budget at TRL 3; later co-located field samples
R4	Estimate silica exposure	RCS estimate = calibrated PM4 mass x site silica fraction, always labeled as an estimate; dust shown alone when no fraction is entered	Met by design	Firmware sketch review
R5	Update often enough to catch tasks	1 s readings, 1 min logged averages, running 8 h time-weighted average (TWA) updated every minute	Met by design; the sensor gives new readings every second	Firmware sketch review
R6	Warn before a limit is reached	Vibration and LED when the projected 8 h RCS TWA exceeds the action level (default 25 µg/m ³) and a repeated alert at the limit (default 50 µg/m ³); both configurable	Met by design and on paper: five scenarios alert as intended, one early warning after a single heavy cut (CAL-001, G). Defaults per DBG-DDR-001 D3	Firmware sketch review
R7	Last a full shift	12 h of continuous sampling per charge	Met on paper with the 2,000 mAh cell (DBG-DDR-002 D8): 17.7 h at the typical 55 mA and 25 °C; 12.8 h at the 65 mA maximum and 0 °C (CAL-001, A)	Power budget calculation
R8	Wearable in the breathing zone	Inlet worn within 30 cm of the nose and mouth; clip or harness loop	Met on paper: inlet 242 mm from the nose and mouth at a collar or upper-strap mount; a mount more than 269 mm below them fails (CAL-001, I)	Massing model
R9	Light and small	Mass 120 g or less; no larger than 75 x 55 x 35 mm	Met on paper: 119.8 g, a 0.2 g margin with the 2,000 mAh cell; 64 x 52 x 33 mm with the clip (CAL-001, J)	Massing model, then weighing
R10	Survive the site	Electronics splash and dust protected (IP54 target except the sensor air path);		

ID	Requirement	Target	Status at TRL 3 (DBG-CAL-001)	Verification (TRL 3 or later)
		inlet facing down; survives a 1.5 m drop onto concrete	Not verifiable at TRL 3: 750 to 1,500 g deceleration on a 1.5 m drop; gasketed joint untested (CAL-001, K)	Design review; later drop and spray tests
R11	Work in site conditions	0 to 45 °C, 10 to 90 % RH non-condensing; readings flagged when humidity or spray may bias them. Use rule: worn shaded when the ambient is above 40 °C in full sun (DBG-DDR-002 D10)	At risk. Under the use rule the shell stays at or below 57.8 °C, 2.2 K under the sensor's 60 °C limit; without it, about 63 °C in full sun at 45 °C. Sensor best performance only to 40 °C and 80 % RH; humidity flag possible with the RH sensor (CAL-001, C)	Literature and datasheet review
R12	Keep a shift log	At least 30 shifts of 1 min records on the badge; export over Bluetooth Low Energy to the worker's phone	Met on paper: 11.5 kB per 12 h shift, about 182 shifts in 2 MB (CAL-001, H)	Storage calculation
R13	Protect the worker's data	Stored on the badge and the worker's phone; shared with an employer only by the worker's choice	Met by design (DBG-DDR-001 D5)	Design review
R14	Safe in hazardous atmospheres	Certified intrinsically safe for gassy mines and explosive atmospheres	Not met and out of scope for this prototype; it must be labeled not for such use	Design review
R15	Low cost and buildable	Parts \$90 or less; no custom PCB required for the first build	Not met. \$91.00 against \$90 with the 2,000 mAh cell; the budget figure is open (DBG-DDR-002 O3) (CAL-001, L)	Priced BOM

■ Assumptions

- The respirable convention in ISO 7708 has a 50 % cut near 4 µm aerodynamic diameter; PM4 from an optical sensor is used as a proxy.
- A site silica fraction (the quartz share of respirable dust) is available for each site and task from at least one filter sample analyzed by a laboratory or a field infrared method. How it is obtained where no laboratory is near is still open (DBG-DDR-001, O2).
- The wearer follows the sun use rule in R11; the badge cannot enforce it.
- Default limits follow US OSHA and MSHA (50 µg/m³ limit and 25 µg/m³ action level, 8 h TWA), decided by Amish on 2026-09-25 (DBG-DDR-001 D3, DBG-DDR-002). Other jurisdictions can be configured, for example the EU binding limit of 100 µg/m³.
- The calculation assumptions (sensor figures, cell, error terms, thermal coefficients) are listed in DBG-CAL-001, Table 1.
- Shifts are up to 12 h; the projected TWA is normalized to 8 h as the regulations define it.